

SDG MAPPING DOCUMENT

Power BI Semantic Model: project

Alignment of E-Commerce Sales Analytics Model to the United Nations
Sustainable Development Goals (SDGs) 2030 Agenda

Document Date:	07 May 2026
Model Name:	project (Power BI Desktop)
Data Domain:	E-Commerce Sales Analytics
SDGs Identified:	SDG 8, SDG 9, SDG 10, SDG 12, SDG 17
Total Records:	50,000 orders 4 regions 6 product categories

1. Executive Summary

This document establishes the formal mapping between the Power BI semantic model titled "project" and the United Nations Sustainable Development Goals (SDGs) under the 2030 Agenda for Sustainable Development. The model tracks e-commerce sales, profitability, pricing, discount behaviour, customer satisfaction, and regional distribution across 50,000 order transactions spanning 2022–2023.

Five SDGs have been identified as directly relevant based on a systematic review of the model's tables, columns, measures, and analytical focus areas:

SDG No.	Goal Name	Primary Connection to Model
SDG 8	Decent Work & Economic Growth	Revenue, profit, order growth, YoY trends — measures economic activity and trade expansion
SDG 9	Industry, Innovation & Infrastructure	Digital commerce platform, electronic payments, multi-region infrastructure
SDG 10	Reduced Inequalities	Regional sales distribution (North America, Asia, Europe, Middle East)
SDG 12	Responsible Consumption & Production	Profit margin, discount discipline, revenue efficiency, cost of goods tracking
SDG 17	Partnerships for the Goals	Cross-regional analytics, multi-category trade, payment diversity enabling inclusive commerce

2. Model Overview

The "project" semantic model is a tabular analytics model built in Power BI Desktop (Compatibility Level 1600). It is centred on a single fact table, `cleaned_dataset`, augmented by a Date dimension table and 65 DAX measures organised across 8 analytical folders.

2.1 Data Structure

Element	Details	SDG Relevance
Primary Table	cleaned_dataset (50,000 rows)	Full commercial transaction record
Date Dimension	Date (2022–2023)	Time-series economic analysis
Measures Table	65 DAX measures across 8 folders	Structured KPI framework
Product Categories	Fashion, Electronics, Beauty, Sports, Books, Home & Kitchen	Consumer production/consumption diversity
Customer Regions	North America, Asia, Europe, Middle East	Geographic equity and inclusiveness
Payment Methods	UPI and others	Financial inclusion infrastructure

2.2 Key Business Metrics (Aggregated)

Metric	Value	SDG Signal
Total Revenue	₹3.29 Crore (32,866,574)	SDG 8 — Economic output
Total Profit	₹16.43 Lakh (1,643,329)	SDG 8, SDG 12 — Profitability
Total Orders	50,000	SDG 8 — Market participation
Profit Margin	5.0%	SDG 12 — Responsible production
Avg Discount	13.3%	SDG 12 — Pricing discipline
Avg Customer Rating	3.0 / 5.0	SDG 8 — Decent service quality
Regions Covered	4 global regions	SDG 10, SDG 17 — Inclusion & partnership

3. Methodology

The SDG mapping was conducted using the following approach:

- Step 1 — Model Extraction: The full semantic model schema was extracted via the Power BI Modeling MCP, including all tables, columns (29 fields), and 65 DAX measures across Sales, Profitability, Pricing, Customer, Products, and Time Intelligence folders.
- Step 2 — Domain Classification: The e-commerce data domain was characterised based on transaction types, geographic scope, product categories, and analytical metrics.
- Step 3 — SDG Screening: All 17 UN SDGs were screened for relevance against identified data elements. SDGs were included where a direct, substantive link existed between model content and SDG targets/indicators.
- Step 4 — Indicator Mapping: Specific SDG targets (e.g., Target 8.2 — higher economic productivity) were matched to individual measures and columns in the model.
- Step 5 — Justification: For each mapping, a rationale was documented explaining how the data element contributes evidence toward the SDG indicator.

SDGs 1–7, 11, 13–16 were reviewed but excluded as the model does not contain data directly measurable against those goals (e.g., poverty rates, health outcomes, climate data, biodiversity, or governance metrics).

4. Detailed SDG Mappings

The following sections provide a detailed breakdown of each selected SDG — the rationale for selection, the specific targets addressed, and a column-by-column / measure-by-measure mapping table.

SDG 8 Decent Work and Economic Growth

4.1 SDG 8 — Why Selected

SDG 8 calls for sustained, inclusive, and sustainable economic growth, full and productive employment, and decent work for all. The project model is fundamentally a commercial performance dashboard that tracks revenue generation, order volume, profitability, and customer engagement — all of which are direct proxies for economic activity and productive commerce.

Relevant SDG 8 Targets addressed by this model:

- Target 8.2 — Higher levels of economic productivity through diversification, technological upgrading and innovation.
- Target 8.3 — Promote development-oriented policies that support productive activities, decent job creation, entrepreneurship, and formalisation of SMEs.
- Target 8.10 — Strengthen the capacity of domestic financial institutions to encourage and expand access to banking, insurance and financial services for all.

Model Element	SDG Indicator / Target	Justification
Total Revenue	Target 8.2 — Economic productivity	Measures overall economic output of the commercial platform across all categories and regions
Total Profit / Profit Margin Pct	Target 8.3 — Productive enterprise sustainability	Tracks 5% net profit margin as a measure of commercial viability and enterprise health
Total Orders (50,000)	Target 8.2 — Volume of productive transactions	Reflects scale of market participation; high order volume indicates active economic engagement
Revenue YoY % / YoY Variance	Target 8.2 — Economic growth rate	Year-over-year growth measures show whether economic activity is expanding or contracting
Revenue MTD / QTD / YTD	Target 8.3 — Regular business cycle activity	Periodic revenue measures indicate steady, formalised commercial operations

Model Element	SDG Indicator / Target	Justification
Avg Rating (3.0/5.0)	Target 8.3 — Decent service quality	Customer satisfaction score reflects quality of work and service delivery in commerce
Payment Method (UPI etc.)	Target 8.10 — Financial inclusion	Digital payment infrastructure supports formal financial participation for consumers
Revenue Rolling 3M / 6M / 12M	Target 8.2 — Sustained growth patterns	Rolling averages expose structural growth trends beyond short-term fluctuations

SDG 9**Industry, Innovation and Infrastructure**

4.2 SDG 9 — Why Selected

SDG 9 promotes inclusive and sustainable industrialisation, fosters innovation, and builds resilient infrastructure. The project model represents a digitally-enabled e-commerce system — itself a form of digital infrastructure — that spans multiple global regions, supports electronic payments, and tracks product innovation through category diversification.

Relevant SDG 9 Targets addressed by this model:

- Target 9.1 — Develop quality, reliable, sustainable and resilient infrastructure to support economic development and human well-being.
- Target 9.3 — Increase access of small-scale industrial and other enterprises to financial services and integration into value chains and markets.
- Target 9.b — Support domestic technology development, research and innovation in developing countries.

Model Element	SDG Indicator / Target	Justification
Product Category (6 types)	Target 9.b — Industrial diversification	6 categories (Fashion, Electronics, Beauty, Sports, Books, Home & Kitchen) reflect a diversified industrial production base
Electronics category	Target 9.1 — Tech infrastructure products	Electronics sales reflect consumer access to technology goods — a proxy for tech infrastructure penetration
Payment Method diversity	Target 9.3 — Financial service infrastructure	Multiple payment channels (UPI, others) indicate accessible, inclusive financial infrastructure
Distinct Products measure	Target 9.b — Product innovation breadth	Number of distinct product IDs (thousands) signals innovation and variety within the supply chain
Order Date (2022–2023)	Target 9.1 — Operational platform resilience	Continuous operations over 2 years demonstrate infrastructure reliability and uptime
4 Global Regions served	Target 9.1 — Cross-regional infrastructure	North America, Asia, Europe, Middle East coverage shows mature digital trade infrastructure

4.3 SDG 10 — Why Selected

SDG 10 aims to reduce inequality within and among countries. The model's regional segmentation across four geographically and economically diverse areas — North America, Asia, Europe, and the Middle East — enables analysis of equitable access to markets, comparative revenue performance across regions, and the distribution of commercial opportunity globally.

Relevant SDG 10 Targets addressed by this model:

- Target 10.2 — Empower and promote the social, economic and political inclusion of all, irrespective of age, sex, disability, race, ethnicity, origin, religion or economic status.
- Target 10.3 — Ensure equal opportunity and reduce inequalities of outcome through equal access to services and markets.
- Target 10.b — Encourage official development assistance and financial flows to states where the need is greatest.

Model Element	SDG Indicator / Target	Justification
Customer Region (4 regions)	Target 10.3 — Geographic market inclusion	Even distribution (~12,500 orders each) across North America, Asia, Europe, Middle East shows equitable market access
Total Revenue by Region	Target 10.2 — Economic equity analysis	Revenue comparability across regions (₹81–83 Lakh each) enables inequality gap measurement
Distinct Regions measure	Target 10.b — Cross-regional commercial flow	Tracks whether all regions are consistently served, highlighting exclusion risks
Rating Band / Avg Rating	Target 10.3 — Service quality equity	Monitoring satisfaction across geographies identifies whether service quality is equal across regions
Revenue by Payment Method	Target 10.2 — Financial access equity	Examining revenue by payment type reveals digital payment adoption disparities across regions
Season / Day Type columns	Target 10.3 — Temporal access patterns	Identifies whether certain consumer groups (weekday vs. weekend) have different purchasing access

4.4 SDG 12 — Why Selected

SDG 12 promotes sustainable consumption and production patterns. The model tracks pricing efficiency, discount behaviour, cost of goods, profit margins, and revenue tiers — all of which are critical to understanding whether commercial activity is sustainable, responsible, and value-preserving for both producers and consumers.

Relevant SDG 12 Targets addressed by this model:

- Target 12.2 — Achieve sustainable management and efficient use of natural resources.
- Target 12.6 — Encourage companies to adopt sustainable practices and to integrate sustainability information into their reporting cycles.
- Target 12.8 — Ensure that people everywhere have relevant information and awareness for sustainable development and lifestyles.

Model Element	SDG Indicator / Target	Justification
Profit Margin Pct (5%)	Target 12.6 — Sustainable commercial operations	A consistent 5% margin signals stable, non-wasteful pricing practices across the supply chain
Discount Percent / Avg Discount Pct (13.3%)	Target 12.2 — Resource/value efficiency	Excessive discounting wastes commercial value; monitoring discount depth ensures pricing sustainability
Discount Impact Pct	Target 12.6 — Transparency in pricing	Quantifies revenue foregone due to promotions — a key sustainability reporting metric for commerce
Cost of Goods column	Target 12.2 — Production cost awareness	Direct cost tracking enables efficiency measurement and reduces waste in production-to-sale pipeline
Revenue Tier / Profit Tier	Target 12.8 — Transparent consumer information	Tiered classifications give consumers and managers clear signals about value positioning
Discount Band / Price Band	Target 12.6 — Reporting sustainability categories	Structured banding supports sustainability-oriented segmentation in annual/periodic reports
Revenue per Unit	Target 12.2 — Per-unit efficiency	Higher revenue per unit indicates less resource

Model Element	SDG Indicator / Target	Justification
		expenditure per unit of economic output
Effective Discount Pct	Target 12.6 — Actual pricing transparency	Actual vs. nominal discount comparison reveals true consumer cost burden and producer efficiency

4.5 SDG 17 — Why Selected

SDG 17 calls for strengthening the means of implementation and revitalising the global partnership for sustainable development, including through trade, data, technology, and capacity building. The model's multi-regional, multi-category, multi-payment-channel design embodies a cross-border commercial partnership ecosystem that generates the type of data needed to measure global trade flows and sustainable partnerships.

Relevant SDG 17 Targets addressed by this model:

- Target 17.10 — Promote a universal, rules-based, open, non-discriminatory and equitable multilateral trading system.
- Target 17.11 — Increase the exports of developing countries, with a view to doubling the least developed countries' share of global exports.
- Target 17.18 — Enhance capacity-building support to developing countries to significantly increase the availability of high-quality, timely and reliable data.
- Target 17.19 — Build on existing initiatives to develop measurements of progress on sustainable development that complement GDP.

Model Element	SDG Indicator / Target	Justification
Customer Region × Revenue	Target 17.10 — Multilateral trade analytics	Cross-regional revenue analysis supports monitoring of equitable trade participation
Product Category × Region	Target 17.11 — Trade diversification	Category-level trade data across 4 regions shows distribution of goods in global commerce
Orders YoY / Revenue YoY	Target 17.19 — GDP-complementary metrics	Order-based economic measures complement traditional GDP by capturing digital-commerce activity
Review Count / Avg Reviews	Target 17.18 — Quality data generation	Consumer review data provides high-frequency, real-time socio-economic signals beyond official statistics
Date Dimension (2022–2023)	Target 17.18 — Timely, reliable data	Two-year longitudinal data enables trend analysis that supports evidence-based partnership decisions
Payment Method diversity	Target 17.10 — Open financial systems	Multiple payment methods indicate open, non-

Model Element	SDG Indicator / Target	Justification
		discriminatory access to trade infrastructure
High Value Review Flag	Target 17.19 — Alternative progress indicators	Binary quality flag introduces non-monetary well-being measurement alongside financial KPIs

5. SDGs Reviewed but Excluded

The following SDGs were systematically reviewed against the model's content but were not selected due to insufficient direct data linkage:

SDG	Goal	Reason for Exclusion
1	No Poverty	No income-level or poverty threshold data in the model
2	Zero Hunger	No food security, agricultural, or nutrition data present
3	Good Health & Well-being	No health, mortality, or wellness metrics available
4	Quality Education	No education attainment or literacy data in scope
5	Gender Equality	No gender or demographic segmentation in the dataset
6	Clean Water & Sanitation	No water or sanitation infrastructure data available
7	Affordable & Clean Energy	No energy consumption or renewables data present
11	Sustainable Cities	No urban planning or city-level geographic data
13	Climate Action	No carbon, emissions, or environmental impact metrics
14	Life Below Water	No marine or aquatic ecosystem data
15	Life on Land	No terrestrial biodiversity or land-use data
16	Peace, Justice & Institutions	No governance, legal, or institutional data tracked

6. Conclusion & Recommendations

The project Power BI model demonstrates a meaningful and multi-layered alignment with five of the seventeen United Nations SDGs. Its strength lies in the depth of its commercial analytics — particularly the 65 DAX measures spanning time intelligence, profitability, and customer behaviour — which enable robust, evidence-based contributions to SDG monitoring.

6.1 Recommendations for Strengthening SDG Alignment

- Add gender / age demographic data to customer records → unlocks SDG 5 (Gender Equality) and deepens SDG 10 (Reduced Inequalities).
- Include carbon/logistics emissions data per shipment → directly enables SDG 13 (Climate Action) and SDG 12 sustainability depth.
- Integrate supplier or vendor location and tier data → strengthens SDG 17 (Partnerships) and SDG 8 (Decent Work) through supply chain visibility.
- Add employee headcount or wage data → provides direct SDG 8 (Decent Work) indicators beyond consumer-side metrics.
- Incorporate city or postal-code level geography → enables SDG 11 (Sustainable Cities) analysis and finer SDG 10 inequality measurement.

6.2 SDG Coverage Summary

SDG	Goal	Coverage Level	Key Model Asset
SDG 8	Decent Work & Economic Growth	HIGH (8 mappings)	Revenue, Profit, YoY measures
SDG 9	Industry, Innovation & Infrastructure	MEDIUM (6 mappings)	Product categories, Payment methods
SDG 10	Reduced Inequalities	MEDIUM (6 mappings)	Regional distribution, Rating equity
SDG 12	Responsible Consumption & Production	HIGH (8 mappings)	Discount, Margin, Cost of Goods
SDG 17	Partnerships for the Goals	MEDIUM (7 mappings)	Cross-region trade, Review data